

Adaptive Vibration Cancellation in SPECULA: implementation, simulation and analysis

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August 26, 2026

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1 The algorithm

The AVC module of Muradore et al. [1] is designed to sit *in parallel* with an existing AO controller and remove one narrowband vibration, without requiring the main loop to be retuned. It targets vibrations above the loop bandwidth, where the rejection transfer function of a classical integrator amplifies rather than attenuates.

1.1 State and control law

Each AVC instance tracks one tone and carries four adapted numbers,

$$\mathbf{x} = (x_1, x_2, x_3, x_4)^\top = (P_R, P_I, \lambda_c, \lambda_s)^\top, \quad (1)$$

interpreted by the paper as the real and imaginary parts of the plant frequency response at the vibration frequency, and the cosine/sine quadratures of the disturbance. It is convenient to group them into two complex numbers,

$$P = x_1 + ix_2 \quad (\text{plant}), \quad \Lambda = x_3 + ix_4 \quad (\text{disturbance}). \quad (2)$$

Three conventions are used throughout. An overbar denotes complex conjugation, $\bar{P} = x_1 - ix_2$; a hat denotes a quantity held by the algorithm as an estimate, $\hat{P}$ and $\hat{\Lambda}$; and a star denotes the corresponding true value of the physical system, $P_\star$ and $\Lambda_\star$. The distinction between $\hat{P}$ and $P_\star$ matters from Sec. 3 onwards, where the question is precisely whether the former can be made to approach the latter. From these the algorithm forms the control quadratures $\vartheta = \theta_c + i\theta_s$, with

$$\theta_c = -\frac{x_1x_3 - x_2x_4}{D}, \quad \theta_s = -\frac{x_1x_4 + x_2x_3}{D}, \quad D = x_1^2 + x_2^2 + \epsilon = |P|^2 + \epsilon, \quad (3)$$

which in complex form is simply

$$\vartheta = -\frac{P\Lambda}{D} \xrightarrow{\epsilon \rightarrow 0} \vartheta = -\frac{\Lambda}{\bar{P}}, \quad (4)$$

the first form following from $P\Lambda = (x_1x_3 - x_2x_4) + i(x_1x_4 + x_2x_3)$, and the limit from $D \rightarrow |P|^2 = P\bar{P}$ as $\epsilon \rightarrow 0$. The algorithm computes the correction $u(t) = \theta_c \cos \alpha + \theta_s \sin \alpha$, where α is the running phase estimate. The idea is to cancel the disturbance Λ by applying a command that is the estimate of the input vibration multiplied by the inverse of the plant response, so that the sum of the command and the disturbance is zero at the sensor.

The regularizer ϵ . The quantity $\epsilon \geq 0$ in (3) requires a note, because it is not part of the published algorithm. The paper divides by $x_1^2 + x_2^2$ exactly, i.e. $\epsilon = 0$. That denominator is the squared magnitude of the *plant estimate*, so the commanded amplitude diverges whenever the estimate approaches the origin and, nothing in the update law prevents it from going to infinity. Implementations therefore guard the division. ESO's *updateAVC.m* freezes ϑ to $(1, 0)$ whenever $x_1^2 + x_2^2 < \theta_{\min}$ with $\theta_{\min} = 10^{-2}$: this is a *discontinuous* clamp, which in our simulations caused the states to jump between the boundary clip values. A smooth equivalent is also implemented, $D = |P|^2 + \epsilon$ with $\epsilon = \theta_{\min}$, which bounds the commanded amplitude by $\sim 1/\epsilon$. In the end, ϵ is a numerical safeguard with an arbitrary value.

1.2 Adaptation

With $c := \cos \alpha$, $s := \sin \alpha$ the algorithm defines the regressor and the prediction error

$$\mathbf{W} = (\theta_c c + \theta_s s, \theta_s c - \theta_c s, c, s)^\top, \quad e = \mathbf{W}^\top \mathbf{x} - y, \quad (5)$$

y being the measured residual of the mode of interest, and updates the state as:

$$\mathbf{x} \leftarrow \mathbf{x} - \mu \mathbf{W} e, \quad \mu = g_x T_s. \quad (6)$$

A separate phase-locked loop tracks the vibration frequency [2, 3], following the direct (PLL-based) scheme of Bodson and Douglas [4]:

$$\omega \leftarrow \omega + 2g_\omega T_s \sin \alpha e, \quad \alpha \leftarrow \alpha + T_s \omega + 2kg_\omega T_s \sin \alpha e. \quad (7)$$

symbol	code name	role
tuning parameters		
f_0	<i>freq</i>	initial guess of the vibration frequency [Hz]
g_x	<i>gx</i>	adaptation gain of the state $\mathbf{x}$
g_ω	<i>gomega</i>	adaptation gain of the frequency loop
k	<i>k</i>	phase/frequency coupling in the PLL
c_x	<i>c</i>	extra gain applied only to the x_1, x_2 update
N	<i>n_oversample</i>	oversampling factor in the phase/frequency update
ϵ	<i>theta_min_energy</i>	division guard in (3)
—	<i>soft_clamp</i>	smooth guard instead of the discontinuous clamp
—	<i>adapt_plant</i>	adapt or freeze x_1, x_2
—	<i>n_avc</i>	number of independent instances (one per tone)
initial conditions		
P_R, P_I	<i>x0, x1</i>	initial plant estimate
α_0	<i>alpha0</i>	initial phase
internal states		
$\mathbf{x}$	—	$(P_R, P_I, \lambda_c, \lambda_s)$
ω, α	—	frequency and phase of the internal oscillator
α_c	—	compensated-summation carry for α

Table 1: Parameters and states of the SPECULA AVC object.

The algorithm performs two online updates: an estimate of the plant response at the vibration frequency (x_1, x_2) and an estimate of the disturbance quadratures (x_3, x_4). In parallel to these, the PLL tracks the vibration frequency and phase. The gains g_x and g_ω set the adaptation rates of the plant/disturbance and the PLL, respectively.

1.3 Implementation notes

The SPECULA *avc* processing object has been defined, to implement (3)–(7) exactly, and is validated against an independent line-by-line translation of ESO’s *updateAVC.m* to 10^{-6} relative. Beyond the division guard already discussed, one point deserves mention.

1.4 Parameters and internal states

Table 1 lists the arguments and internal states of the SPECULA AVC object.

c_x constant In the code this constant is called c ; we rename it c_x here only to avoid a clash with $c := \cos \alpha$ used in (5). It multiplies the first two components of the update

$$x_{1,2} \leftarrow x_{1,2} - \mu c_x W_{1,2} e, \quad x_{3,4} \leftarrow x_{3,4} - \mu W_{3,4} e, \quad (8)$$

so it sets the rate of the plant half of the state relative to the disturbance half. This turns out to be necessary: the plant response is not identifiable from a single tone, so the plant estimate drifts and the loop diverges if it is adapted quickly. In practice, $c_x = 0$ in most applications. This will be discussed in Sec. 3.

α_c is the numerical correction. The phase is accumulated once per frame, $\alpha \leftarrow \alpha + T_s \omega + \dots$, at 1 kHz for many seconds. In single precision this running sum loses low-order bits, and the accumulated phase error would appear as a slow, spurious frequency offset. The object therefore carries a Kahan compensated-summation term [5, 6] α_c , which holds the part of each increment lost to rounding and reinjects it at the next step. (The implementation is the classical Kahan recursion; the Kahan–Babuška–Neumaier variant [7] additionally compensates when the

block	parameter	value	note
Telescope / sampling	pupil diameter	1.0 m	$64 \times 0.015625 \text{ m}$
	pupil sampling	$64 \times 64 \text{ px}$	
	loop rate	1 kHz	$T_s = 1 \text{ ms}$
	run length	10 s	10^4 steps
Atmosphere	seeing	0.6 arcsec	single layer
	outer scale L_0	25 m	
	layer height	0 m	C_n^2 weight 1.0
	wind speed / direction	5 m/s / 10 deg	frozen flow
	phase screen	1024 px	seed = 1 (reproducible)
Source	magnitude	5	on-axis NGS
	wavelength	600 nm	
WFS (Shack–Hartmann)	subapertures	8×8	
	pixels per subaperture	8×8	
	pixel scale	0.5 arcsec/px	FoV 4 arcsec
	wavelength	600 nm	
Detector	size	$64 \times 64 \text{ px}$	integration 1 ms
	bandwidth	300 nm	
	quantum efficiency	0.3	
	photon noise	on	
	readout noise	on	$1.0 \text{ e}^-/\text{px}/\text{frame}$
Reconstruction	modes	40 Zernike	<i>scao_rec_n8_th0.5</i>
Main control	integrator gain	0.5 (all modes)	
	delay	2 frames	+1 frame from propagation
	actuator low-pass	80 Hz, 1st order	first-order Butterworth
	DM	40 Zernike modes	obstruction ratio 0.1
Vibration	amplitude	120 nm	on mode 0 (tip)
	frequency	47 Hz	

Table 2: Simulation parameters. All runs share the atmospheric seed, so configurations differ only in the AVC settings.

increment exceeds the accumulator, which does not arise here.) It is a pure numerical device: it has no control-theoretic meaning, does not appear in the paper’s equations, and is present only so that the phase accumulator behaves as if computed in higher precision.

2 Simulation in SPECULA

2.1 Pipeline

The demonstration is a complete single-conjugate AO loop. Two features are worth highlighting. First, the vibration is injected as a *separate* deformable mirror driven by a sinusoidal generator, so it enters the pupil independently of the atmosphere and of the loop’s own correction, as a real mechanical vibration would. Second, AVC runs in parallel with the main integrator: its command is summed with the integrator’s before the actuator model, so it can be switched on and off without touching the main controller.

2.2 Why 47 Hz is the interesting place to put a vibration

The integrator rejects low frequencies well but, above its bandwidth, the rejection transfer function $|S| = |1/(1 + H_{OL})|$ rises above unity. For this loop the measured rejection is strongly *positive* at 47 Hz: the 120 nm injected tone emerges as a 891 nm residual, an amplification factor

	baseline (no AVC)	with AVC	change
mode-0 (tip) residual RMS	630.4 nm	26.5 nm	−95.8%
47 Hz line amplitude	891 nm	0.6 nm	−63.0 dB
frequency estimate	—	46.96 Hz	from a 45 Hz guess

Table 3: Performance of the recommended configuration, 10 s run, statistics over the second half.

of 7.4 (+17.4 dB). The vibration is injected only in mode 0, for simplicity, while the integrator compensates for the first 40 modes. The main controller is actively making this disturbance worse, and cannot be retuned to fix it without giving up low-frequency performance.

2.3 Calibrating the plant

AVC needs two numbers before it can run: the frequency of the vibration, and the response of the loop to its own command at that frequency. Both come from an offline calibration in two stages, the first feeding the second.

Stage 1: find the tone. Run the loop *without* AVC, record the residual of the mode of interest, and take its power spectrum. Feeding that PSD to *specula/lib/find_vib_peaks.py* returns the vibration peaks: the routine estimates a smoothed noise floor, flags samples standing above it, groups them, and fits each group with a second-order (AR2) model to extract the frequency, damping and integrated power.

Stage 2: measure the response at that frequency. The frequency from stage 1 is then used to calibrate the frequency response of the plant P at that frequency. The calibration is performed in closed-loop, with the integrator running, by injecting a known sinusoid into the AVC command path and measuring the response at the sensor. For this calibration to be valid, the residuals must be small enough that the Shack–Hartmann sensor is linear, so that the vibration frequency is not moved and the response is not saturated.

Status in this work. In the simulations reported here stage 1 was bypassed, since the vibration was directly injected at $f_0 = 47\text{ Hz}$. The peak finding routine is implemented and unit-tested but is not yet called by the demo, and connecting the two stages into a single calibration script remains to be done.

2.4 Best configuration and results

The best-performing configuration adapts the frequency and the disturbance quadratures, and holds the plant at its measured value:

parameter	value	role
<i>freq</i>	45 Hz	initial guess (deliberately 2 Hz off)
<i>gx</i>	3.0	disturbance adaptation rate
<i>gomega</i>	0.2	PLL rate (re-tune with tone SNR)
<i>x0</i> , <i>x1</i>	−7.1027, +1.3137	measured closed-loop plant
<i>adapt_plant</i>	<i>false</i>	freeze x_1, x_2
<i>soft_clamp</i>	<i>true</i>	smooth division guard

The vibration is essentially eliminated. The frequency loop converges from a wrong initial guess and holds to within 0.04 Hz. Convergence takes roughly 3 s, set by the disturbance adaptation rate g_x ; the frequency lock is faster.

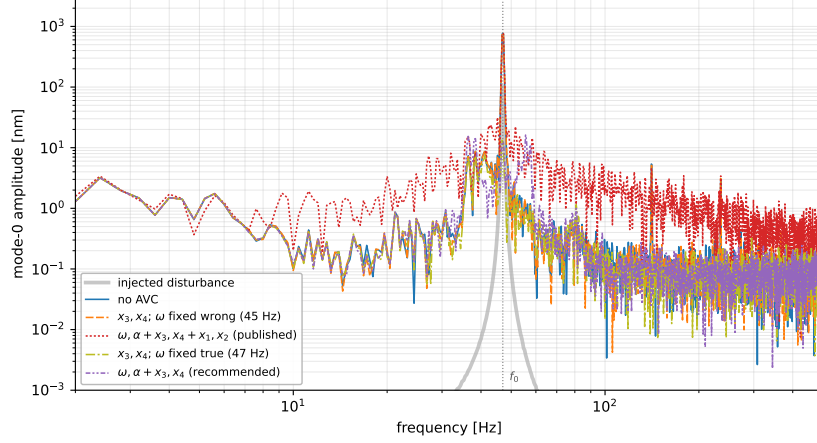


Figure 1: Steady-state mode-0 residual spectra, with the injected disturbance for reference (grey). Without AVC the loop amplifies the tone to 17.4 dB above the input. The two failing configurations track that peak exactly; the two working ones remove it, ending 18.2 dB *below* the input. In the original configuration, adapting both the plant and the disturbance estimate, a broadband amplification is generated at all frequencies due to a wrong phasing of the command.

3 Analysis: what works, and what cannot

The configuration above deliberately switches off the plant adaption. This section analyzes why.

3.1 What the measurement contains

The algorithm models the measured residual as a sinusoid at the tracked frequency,

$$y(t) = \text{Re}(Z) \cos \alpha(t) + \text{Im}(Z) \sin \alpha(t), \quad Z = \vartheta \overline{P_\star} + \Lambda_\star = \Lambda_\star - \kappa \hat{\Lambda}, \quad \kappa = \overline{\left(\frac{P_\star}{\hat{P}} \right)}, \quad (9)$$

with $P_\star, \Lambda_\star$ the true plant and disturbance and ϑ the applied control.

Once $\alpha(t)$ is known, the measurement depends on the unknown parameters only through the complex number Z , i.e. through two real numbers. Nor does it improve with observation time. The regressor has the form $\mathbf{W} = \cos \alpha \mathbf{u}_1 + \sin \alpha \mathbf{u}_2$ with $\mathbf{u}_{1,2}$ independent of phase (Sec. 3.5), so $\sum_t \mathbf{W} \mathbf{W}^\top$ has rank two for any number of samples (length of the experiment).

The whole experiment therefore yields two real numbers about the static parameters, however long it runs. Every identifiability question reduces to asking which parameters are determined by Z .

3.2 Why plant identification is ill-posed

Treat the true $(P_\star, \Lambda_\star)$ as jointly unknown, with the applied command ϑ known. Then $Z = \vartheta \overline{P_\star} + \Lambda_\star$ is a linear map $\mathbb{R}^4 \rightarrow \mathbb{R}^2$ both conjugation and multiplication by ϑ are $\mathbb{R}$ -linear, so its kernel is exactly two-dimensional:

$$\delta Z = \vartheta \delta \overline{P} + \delta \Lambda = 0 \quad \Longleftrightarrow \quad \delta \Lambda = -\vartheta \delta \overline{P}. \quad (10)$$

Any change in the plant can be exactly absorbed by a change in the disturbance, leaving every measurement unchanged at every phase, for all time.

In practice, $\hat{P}$ drifts along this kernel. It does not merely fail to converge: $\hat{P}$ sits in the denominator of (3), so the drift feeds back into the control and the loop diverges. In simulation the plant estimate wanders over three decades in magnitude and the full $\pm 180^\circ$ in phase. P

is however not *needed* as an estimate. Two things follow. A cancelling solution exists for any plant estimate. From (9), the plant appears inside the ratio κ , and setting $Z = 0$ gives

$$\hat{\Lambda}_\infty = \Lambda_\star / \kappa, \quad (11)$$

which is well-defined for any $\kappa \neq 0$. Perfect cancellation is then possible whatever finite non-zero $\hat{P}$ is used. Whether the loop reaches that solution, and what happens to $\hat{P}$ while it does, is a question about dynamics rather than about (9). It is treated in Sec. 3.4. Note already that by (11) a magnitude error in $\hat{P}$ does not degrade the steady state: it rescales the converged $\hat{\Lambda}$ and changes the convergence rate, but still yields $Z = 0$. What AVC needs from the plant is a phase, within a generous margin.

3.3 Cost Function Analysis

The minimized quantity $J = \frac{1}{2}(\mathbf{W}^\top \mathbf{x} - y)^2$ is an *equation error*: it measures whether the model is consistent with itself, not whether the vibration has been cancelled. The model, however, is self-consistent by construction.

Substituting the control law (4), $\vartheta = -P\Lambda/D$ with $D = |P|^2 + \epsilon$, into the prediction gives, in the notation of (9),

$$\mathbf{W}^\top \hat{\mathbf{x}} = \text{Re}(\hat{Z}) \cos \alpha + \text{Im}(\hat{Z}) \sin \alpha, \quad \hat{Z} = \vartheta \bar{\hat{P}} + \hat{\Lambda} = -\frac{|\hat{P}|^2}{D} \hat{\Lambda} + \hat{\Lambda} = \hat{\Lambda} \frac{\epsilon}{D} \xrightarrow{\epsilon \rightarrow 0} 0, \quad (12)$$

Because ϑ is built to cancel $\hat{\Lambda}$, the model always predicts perfect cancellation of its own estimate, whatever that estimate is. With the published $\epsilon = 0$ the prediction is identically zero, so

$$J = \frac{1}{2}(0 - y)^2 = \frac{1}{2}y^2, \quad (13)$$

which does not depend on $\mathbf{x}$ at all. A cost independent of its parameters has zero gradient. Notice that y depends on the estimate through the loop, but the equation error definition does not involve dynamics, so the instantaneous gradient is computed holding y fixed, and y is invariant on the current estimate.

The update (6) is therefore not a gradient descent on the cost it nominally minimizes. Holding $\mathbf{W}$ fixed while differentiating is the defining step of the classical *equation-error* or *pseudo-linear regression* family: if $\mathbf{W}$ really were independent of $\mathbf{x}$, then $\mathbf{W}e$ is the exact gradient and (6) is textbook LMS [8], with its convergence theory. The algorithm can be justified by analyzing the dynamics it generates, not by appealing to the cost it minimizes, through (14) below, and the stability proofs cited by the paper [9, 10]. Since $e \simeq -y$, the disturbance part of (6) becomes, after averaging over a cycle,

$$\dot{\hat{\Lambda}} \propto Z = \Lambda_\star - \kappa \hat{\Lambda}, \quad (14)$$

a first-order integral controller on the residual. What makes the algorithm effective is therefore integral action on the measured residual, which is a control objective, rather than the minimization of (12).

A corollary: $\hat{\Lambda}$ converges to the command that cancels the disturbance, not to the disturbance itself. Across runs with different frozen $\hat{P}$ the converged $|\hat{\Lambda}|$ scales with $|\hat{P}|$ to about 1%, as $\hat{\Lambda}_\infty = \Lambda_\star / \kappa$ requires, while all of them cancel equally well. It is a controller state, not an estimate.

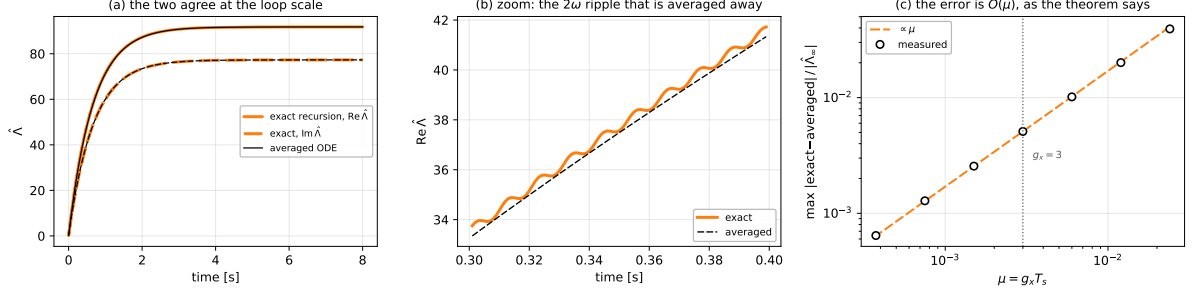


Figure 2: The averaging step. (a) Exact per-sample recursion against the solution of the averaged ODE (18), at $g_x = 3$. (b) Zoom: the 2ω ripple that the average removes. (c) Maximum discrepancy against the small parameter $\mu = g_x T_s$, confirming the $O(\mu)$ error bound.

3.4 Convergence analysis

The two halves of the state converge in different ways. Equation (6) applied to (5), with the plant half carrying its own gain c_x (Sec. 1.4), reads

$$\begin{aligned} x_1 &\leftarrow x_1 - \mu c_x (\theta_c c + \theta_s s) e, & x_3 &\leftarrow x_3 - \mu c e, \\ x_2 &\leftarrow x_2 - \mu c_x (\theta_s c - \theta_c s) e, & x_4 &\leftarrow x_4 - \mu s e. \end{aligned} \quad (15)$$

Grouping as in Sec. 1, $\hat{P} = x_1 + ix_2$ and $\hat{\Lambda} = x_3 + ix_4$, the right-hand pair gives $\delta x_3 + i \delta x_4 = -\mu(c + is)e$, and the left-hand pair uses the identity $W_1 + iW_2 = \vartheta e^{-i\alpha}$. Hence

$$\delta \hat{\Lambda} = -\mu e^{i\alpha} e, \quad \delta \hat{P} = -\mu c_x \vartheta e^{-i\alpha} e. \quad (16)$$

The two are identical except for the gain c_x and the factor $\vartheta e^{-2i\alpha}$. The measured residual can be written equivalently as

$$y(t) = \text{Re}(Z) \cos \alpha + \text{Im}(Z) \sin \alpha = \text{Re}(Z e^{-i\alpha}), \quad (17)$$

Recall also $e \simeq -y$, since the prediction term is $\mathcal{O}(\epsilon)$ by (12). Substituting (17) into (16) and averaging over one cycle, using $\langle e^{\pm 2i\alpha} \rangle = 0$:

$$\langle e^{i\alpha} y \rangle = \frac{1}{2} Z, \quad \langle e^{-i\alpha} y \rangle = \frac{1}{2} \bar{Z}, \quad \implies \quad \dot{\hat{\Lambda}} = \gamma Z, \quad \dot{\hat{P}} = \gamma_p \vartheta \bar{Z}, \quad (18)$$

with $\gamma, \gamma_p > 0$.

Averaging is an approximation, and the error of such approximation is known: the averaged solution departs from the exact recursion by $O(\mu)$ on timescales $O(1/\mu)$ [11, 12], the small parameter being the per-sample step $\mu = g_x T_s$. Figure 2 confirms both the size (0.5% of $|\hat{\Lambda}_\infty|$ at $g_x = 3$) and the linear scaling. What licenses the stability analysis that follows is the second half of the averaging theorem, stating that hyperbolic equilibria of the averaged system are hyperbolic periodic orbits of the exact one, with the same stability type. So, $\hat{\Lambda}$ integrates the residual, while $\hat{P}$ has a nonlinear dynamics lead by

$$\vartheta = -\frac{\hat{\Lambda}}{\hat{P}}.$$

We can study dynamic convergence of the two dynamical subsystems. Starting from Λ , we can expand as follows

$$\dot{\hat{\Lambda}} = \gamma (\vartheta \bar{P}_\star + \Lambda_\star). \quad (19)$$

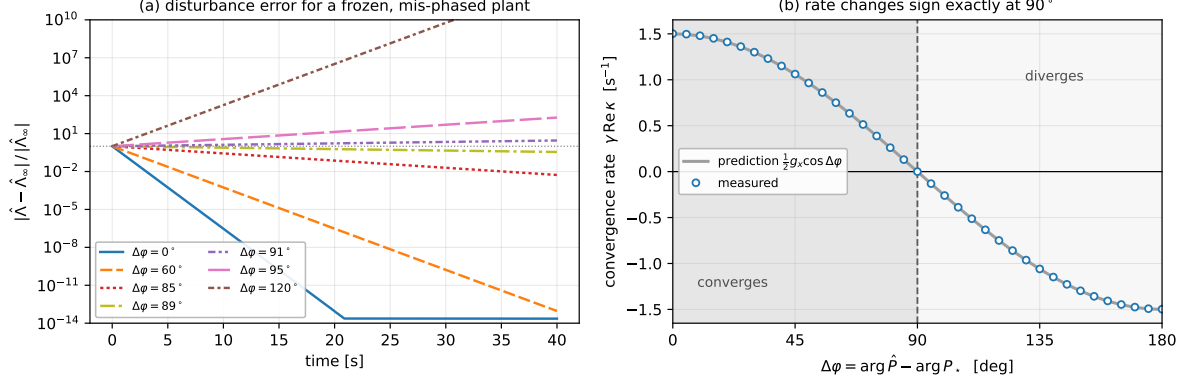


Figure 3: Violation of the margin condition (24), with the plant estimate frozen at a controlled phase error $\Delta\varphi$ and $|\kappa| = 1$. (a) Normalised disturbance error $|\hat{\Lambda} - \hat{\Lambda}_\infty|$: exponential decay for $\Delta\varphi < 90^\circ$ (solid), exponential growth for $\Delta\varphi > 90^\circ$ (dashed); (b) Measured rate against the prediction $\gamma \text{Re } \kappa = \frac{1}{2}g_x \cos \Delta\varphi$.

Recalling the definition of ϑ , we can write

$$\dot{\hat{\Lambda}} = \gamma \left(-\frac{\hat{\Lambda}}{\hat{P}} \overline{P_\star} + \Lambda_\star \right) = -\gamma \frac{\overline{P_\star}}{\hat{P}} \hat{\Lambda} + \gamma \Lambda_\star. \quad (20)$$

This is an integrator dynamics. Introducing the complex loop gain

$$\kappa = \frac{\overline{P_\star}}{\hat{P}} = \overline{\left(\frac{P_\star}{\hat{P}} \right)}, \quad (21)$$

equation (20) becomes $\dot{\hat{\Lambda}} = -\gamma\kappa(\hat{\Lambda} - \hat{\Lambda}_\infty)$ with the fixed point

$$\hat{\Lambda}_\infty = \frac{\Lambda_\star}{\kappa} = \frac{\overline{\hat{P}}}{\overline{P_\star}} \Lambda_\star. \quad (22)$$

This is a complex linear system with the single eigenvalue $-\gamma\kappa$, so the error $\delta = \hat{\Lambda} - \hat{\Lambda}_\infty$ evolves as $\delta(t) = e^{-\gamma\kappa t} \delta(0)$, i.e. (the factor $\frac{1}{2}$ in (18) fixing $\gamma = g_x/2$)

$$|\delta(t)| = e^{-\gamma \text{Re } \kappa t} |\delta(0)|, \quad (23)$$

Convergence therefore requires $\text{Re } \kappa > 0$, which by (21) is a condition on the plant phase alone:

$$|\arg P_\star - \arg \hat{P}| < 90^\circ. \quad (24)$$

In continuous time a plant magnitude error of a factor ten is harmless and only slows convergence, whereas a phase error of 91° causes instability. Figure 3 shows this results through the SPECULA simulation. The component-wise update (15) is run with the plant half frozen at $\hat{P} = |P_\star|e^{i(\arg P_\star + \Delta\varphi)}$, so that $|\kappa| = 1$ and the phase error is the only variable, on a single noiseless vibration.

In discrete time the magnitude does re-enter the stability condition. The exact one-sample increment is $\delta\hat{\Lambda} = -\frac{1}{2}\mu(\kappa\delta + \bar{\kappa}\bar{\delta}e^{2i\alpha})$ with $\mu = g_x T_s$ as in (6). The stability limit is set by the worst phase of the cycle, at which the two terms align and the sample gain becomes $1 - \mu\kappa$. Requiring that to lie in the unit disk gives

$$\mu < \frac{2 \text{Re } \kappa}{|\kappa|^2}. \quad (25)$$

adapted parameters	mode-0 RMS	47 Hz amp.	vs. baseline	PSD vs. input
none (no AVC)	629.8 nm	890.3 nm	—	+17.4 dB
x_3, x_4 ; ω fixed <i>wrong</i> (45 Hz)	646.2 nm	913.6 nm	+0.2 dB	+17.6 dB
$\omega, \alpha + x_3, x_4 + x_1, x_2$ (as published)	634.8 nm	892.6 nm	+0.0 dB	+17.4 dB
x_3, x_4 ; ω fixed <i>true</i> (47 Hz)	20.7 nm	16.7 nm	−34.5 dB	−16.1 dB
$\omega, \alpha + x_3, x_4$ (recommended)	26.1 nm	12.1 nm	−37.4 dB	−18.2 dB

Table 4: Ablation over the adapted parameter set.

It is worth noting that the limit is $\hat{\Lambda}_\infty \neq \Lambda_\star$ unless $\hat{P} = P_\star$. What $\hat{\Lambda}$ converges to is whatever value cancels the tone *given* the plant estimate in use, since $\vartheta_\infty = -\hat{\Lambda}_\infty/\hat{P} = -\Lambda_\star/\overline{P}_\star = \vartheta_\star$. The disturbance is not identified, but the canceling command is. For the control problem that is sufficient, and it is why an approximate calibration of $\hat{P}$ still gives full rejection (Sec. 2.4). Notice that the convergence of the $\hat{\Lambda}$ dynamics implies convergence of the plant estimate $\hat{P}$ dynamics, since ϑ converges to $\vartheta_\star$ and Z converges to zero. However, the plant estimate $\hat{P}$ does not converge to the true plant $P_\star$, but rather drifts along a manifold of equivalent solutions that yield the same control action. Also, existence of a stationary point does not guarantee stability. The problem of the neutral stability of the plant estimate is that, if the plant estimate drifts too far from the true plant, the convergence condition (24) may be violated, leading to divergence of the disturbance estimate and ultimately to instability of the control loop.

3.5 Are the regressors persistently exciting?

This is the classical way in system identification theory to phrase the analysis. Persistent excitation requires the averaged information matrix $\langle \mathbf{W}\mathbf{W}^\top \rangle$ to be positive definite. Writing $\mathbf{W} = c\mathbf{u}_1 + s\mathbf{u}_2$ with $\mathbf{u}_1 = (\theta_c, \theta_s, 1, 0)^\top$ and $\mathbf{u}_2 = (\theta_s, -\theta_c, 0, 1)^\top$, and averaging over a cycle ($\langle c^2 \rangle = \langle s^2 \rangle = \frac{1}{2}$, $\langle cs \rangle = 0$),

$$\langle \mathbf{W}\mathbf{W}^\top \rangle = \frac{1}{2}(\mathbf{u}_1\mathbf{u}_1^\top + \mathbf{u}_2\mathbf{u}_2^\top) = \frac{1}{2}(|\vartheta|^2 + 1)\Pi, \quad (26)$$

with Π the projector onto $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$. The spectrum is $\{\lambda, \lambda, 0, 0\}$ with $\lambda = \frac{1}{2}(|\vartheta|^2 + 1)$, the information matrix is of rank 2, and the regressors are not persistently exciting of order 4.

This is the textbook statement that **a single sinusoid is persistently exciting of order 2** [13, 14]. It supports the estimation of exactly two parameters; four would require excitation of order 4, that is at least two distinct frequencies. It is worth stressing that (26) is not a consequence of the particular $\mathbf{W}$ of (5): *no* choice of four regressors can do better, as long as the model output is a single tone.

3.6 Experimental confirmation

Table 4 tests these conclusions by ablating which parameters are adapted. All configurations share the loop, the atmosphere and the vibration; only the AVC settings differ.

4 A simplified and formally sound AVC algorithm

The analysis of Sec. 3 justifies the formulation of a simpler algorithm that removes the plant adaptation and the optimization approach, based on the ill-posed cost function. The algorithm is a PLL for the vibration frequency and phase and an integrator on the measured residual, with a fixed map from the integrator to the control quadratures, given by the off-line calibration. Because the computation of the control quadratures is not related to x_1 and x_2 anymore, the safeguards on the plant estimate are not needed, so $\epsilon = 0$ and $\theta_{\min} = 0$. What is left is simpler

than the original algorithm, performs identically (under suitable operating conditions), and admits a complete stability statement.

4.1 What is removed

The plant adaptation. The states x_1, x_2 are unobservable from a single tone (Sec. 3.2), contribute two zero eigenvalues to the closed loop (Sec. 3.4), and under noise perform an unbounded random walk that may eventually carry $\arg \hat{P}$ out of the stability cone (24). They are replaced by the closed-loop measurement of Sec. 2.3, performed once, offline.

The regulariser and the division guard. With $\hat{P}$ fixed, the denominator $|\hat{P}|^2$ is a calibration constant rather than a state. It cannot approach zero, so ϵ , the clamp threshold $\theta_{\min}$ and the chattering it caused all disappear. In fact no division is performed at run time at all: the map from (x_3, x_4) to the control quadratures is a fixed 2×2 matrix, evaluated once at configuration time.

The cost function. Section 3.3 shows that $J = \frac{1}{2}e^2$ is independent of the state it is supposed to minimise. Setting $\epsilon = 0$ makes this exact: the prediction term vanishes identically,

$$\mathbf{W}^\top \mathbf{x} \equiv 0 \quad \implies \quad e = -y, \quad (27)$$

verified to 10^{-14} over 2×10^4 random states. There is therefore nothing to be gained by presenting the update as a gradient step. In practice, the $\hat{\Lambda}$ dynamics is a simple integrator on the measured residual, and the plant estimate is not updated at all. There is no need to compute a cost, a gradient, or a prediction.

4.2 The algorithm

The state reduces to the frequency loop (ω, α) and the disturbance quadratures $\hat{\Lambda} = x_3 + ix_4$. The calibrated plant enters through the two constants $\rho = |\hat{P}|$ and $\psi = \arg \hat{P}$, and

$$\vartheta = -\frac{\hat{\Lambda}}{\hat{P}} = -\frac{1}{\rho} e^{i\psi} \hat{\Lambda} \quad \iff \quad \begin{pmatrix} \theta_c \\ \theta_s \end{pmatrix} = -\frac{1}{\rho} \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix}, \quad (28)$$

a fixed rotation by the plant phase followed by a scaling by the inverse plant gain. Each iteration then consists of four operations, with $c = \cos \alpha$, $s = \sin \alpha$ and $\mu = g_x T_s$:

1. **command** $u = \theta_c c + \theta_s s$, added to the output of the main controller;
2. **measure** y , the residual of the mode carrying the vibration;
3. **integrate** $x_3 \leftarrow x_3 + \mu c y$, $x_4 \leftarrow x_4 + \mu s y$;
4. **track** $\omega \leftarrow \omega - 2g_\omega T_s s y$, $\alpha \leftarrow \alpha + T_s \omega - 2kg_\omega T_s s y$;

followed by (28) to refresh ϑ . In complex form the whole of step 3 is

$$\hat{\Lambda} \leftarrow \hat{\Lambda} + \mu e^{i\alpha} y. \quad (29)$$

4.3 What can be guaranteed

Because $\hat{P}$ is now a constant, the averaged dynamics of Sec. 3.4 is a *linear time-invariant* complex system,

$$\dot{\hat{\Lambda}} = \gamma(\Lambda_\star - \kappa \hat{\Lambda}), \quad \gamma = \frac{1}{2}g_x, \quad \kappa = \overline{(P_\star / \hat{P})}, \quad (30)$$

so its properties are global rather than local, and all of them are explicit:

- a **unique** equilibrium $\hat{\Lambda}_\infty = \Lambda_\star / \kappa$, at which the tone is cancelled exactly;

- **global exponential stability** if and only if $|\arg P_\star - \arg \hat{P}| < 90^\circ$;
- a **closed-form rate**, $\frac{1}{2}g_x \operatorname{Re} \kappa$, confirmed from simulation in Fig. 3;
- an explicit **step-size limit** (25), which is loose in practice (much bigger than actually used).

The stability condition is a statement about a quantity that is measured, not adapted, so it can be checked before the loop is closed. The plant *magnitude* does not enter it at all, and the phase has a $\pm 90^\circ$ tolerance.

Read through the internal model principle [15], step 3 places a pair of poles on the imaginary axis at $\pm i\omega$ in the stationary frame, which is exactly the internal model of a sinusoid; asymptotically exact rejection follows. The frequency loop is still nonlinear and adaptive, so (30) describes the behaviour at lock, not the acquisition transient; and what converges is the residual, not an identification of the disturbance, since $\hat{\Lambda}_\infty = \Lambda_\star$ only when the calibration is exact (Sec. 3.4).

4.4 What the algorithm actually is

The simplified algorithm of Sec. 4 is a filtered-X LMS (FxLMS), with the reference signal produced by a phase-locked loop instead of being measured. In the control literature the same algorithm is called adaptive feedforward cancellation (AFC). This consideration allows framing the stability condition (24) in the context of established results for the FxLMS. FxLMS adapts a filter $\mathbf{w}$ using a reference signal $\mathbf{x}$. Before the reference is used in the update, it is filtered through an estimate $\hat{S}$ of the plant between the controller output and the sensor, which in that literature is called the secondary path:

$$\mathbf{w} \leftarrow \mathbf{w} - \mu (\hat{S} * \mathbf{x}) e. \quad (31)$$

A sinusoid passing through any linear filter comes out as a sinusoid at the same frequency, with only its amplitude and its phase changed, so at that one frequency the whole filter is described by two numbers. Writing

$$u = w_c \cos \alpha + w_s \sin \alpha \quad (32)$$

covers every amplitude and every phase, so (w_c, w_s) is already a complete parameterisation. In our algorithm these two weights are x_3 and x_4 , and the delayed copies they multiply are the last two entries $(\cos \alpha, \sin \alpha)$ of the regressor (5). Filtering a sinusoid through $\hat{S}$ does two things: it rotates it by $\arg \hat{S}$, and it scales it by $|\hat{S}|$. Our update does not filter the regressor. It uses $(\cos \alpha, \sin \alpha)$ directly, and applies the rotation and the scaling at the output instead, through the fixed map (28). The correspondence is exact, term by term:

FxLMS	this report
secondary path S	closed-loop plant $P_\star$
its estimate $\hat{S}$	the calibration $\hat{P}$
filtering of the reference	the fixed map (28)
two-tap weight vector	$\hat{\Lambda} = (x_3, x_4)$
step size μ	$g_x T_s$

In a standard FxLMS system the reference signal is measured. Here, the reference is generated internally by the frequency loop. The PLL is a nonlinear oscillator that locks to the vibration, and its output is the reference for the disturbance integrator. The condition (24) is the standard convergence condition for FxLMS. The requirement is that the estimated secondary-path phase be within 90° of the true one, and it is a well known result in active noise control [16, 17, 18]. It is normally stated under the assumption of slow adaptation, which is the same assumption we used when averaging in Sec. 3.4. The resonant-controller reading given above is also standard.

	no AVC	<i>AVCSimplified</i>	change
mode-0 (tip) residual RMS	630.1 nm	26.8 nm	−95.7%
47 Hz line amplitude	891.8 nm	3.7 nm	−47.6 dB
frequency estimate	—	46.96 Hz	from a 45 Hz guess

Table 5: Simplified AVC in the full loop, 10 s run, statistics over the second half. Holding the frequency fixed at the true 47 Hz instead of tracking it brings the residual down to 15.0 nm and the line to 2.8 nm: the small penalty of the running PLL is the price of not needing to know the frequency in advance.

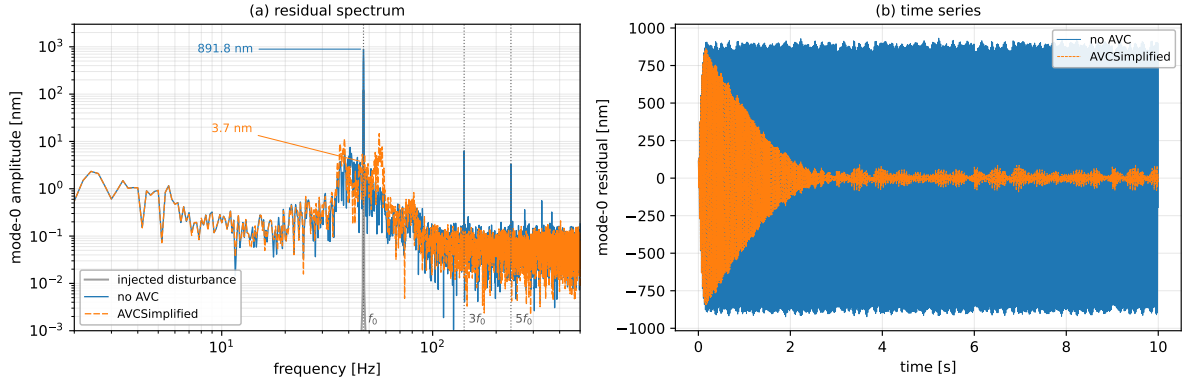


Figure 4: Simplified AVC in closed loop. (a) Steady-state mode-0 spectrum with the module off and on, against the injected disturbance (grey). (b) The same run in the time domain; convergence takes about 2 s, set by $\frac{1}{2} g_x \text{Re } \kappa$.

Messner and Bodson [19] showed that adaptive feedforward cancellation with a sinusoidal regressor is equivalent to a linear time-invariant controller built on the internal model principle. In their formulation the phase advance applied to the regressor plays the role of our $\arg \hat{P}$. The analysis of Sec. 3 then shows that the published algorithm differs from FxLMS in one aspect: it adapts the secondary-path estimate online, which FxLMS practice deliberately avoids. That single difference is what makes it fail, and Sec. 4 recovers the standard algorithm by removing it.

4.5 Simulation results

The object was run in the same closed loop as Sec. 2.4 (*params_scao_avc_simplified.yml*), against the same atmosphere and the same 120 nm tone at 47 Hz, with the plant held at the calibrated $|\hat{P}| = 7.223$, $\arg \hat{P} = +169.52^\circ$ and the frequency loop running from a 45 Hz initial guess.

Figure 4 shows the spectra. The loop without AVC amplifies the injected 120 nm tone to 892 nm, the 17.4 dB waterbed penalty of Sec. 2.2; the module removes it, ending well below the injected level, and leaves the rest of the spectrum untouched; there is no broadband penalty of the kind the plant-adapting configuration produces (Fig. 1).

Harmonics rejection. Figure 4(a) also shows how the AVC removes the odd harmonics of the vibration, for which there is no explicit rejection loop. Some results are collected in Table 6. The odd harmonics have to be generated by an odd-symmetric nonlinearity in the loop. A reasonable guess for the generation of the odd harmonics is the WFS. Since the harmonics are generated by the nonlinearity in the sensor, the AVC does not need to be tuned to reject them: the rejection of the fundamental is enough to remove the harmonics as well.

harmonic	f [Hz]	injected	no AVC	with AVC
1	47	120 nm	891.8 nm	3.69 nm
3	141	—	6.36 nm	0.11 nm
5	235	—	3.33 nm	0.09 nm
7	329	—	0.56 nm	0.10 nm

Table 6: Rejection of Odd Harmonics with AVC on the fundamental

$\Delta\varphi$	mode-0 RMS	47 Hz amp.	behavior
-80°	308 nm	431 nm	converges, slowly
-68°	131 nm	174 nm	converges
-43°	17.5 nm	9.2 nm	converges
0°	14.8 nm	2.8 nm	converges
$+47^\circ$	15.0 nm	4.8 nm	converges
$+77^\circ$	240 nm	320 nm	converges, slowly
$+87^\circ$	951 nm	1284 nm	marginal
$+97^\circ$	1380 nm	1882 nm	diverges
$+137^\circ$	1269 nm	1662 nm	diverges

Table 7: Sweeping the calibration phase

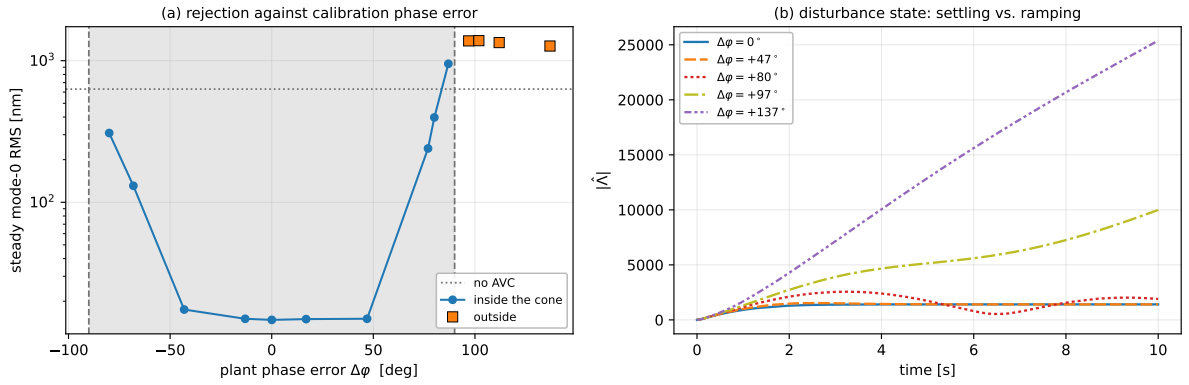


Figure 5: The margin condition in the full simulation. (a) Steady mode-0 RMS against calibration phase error; the shaded band is the predicted cone (24) and the dotted line the residual without AVC. (b) Trajectories of $|\hat{\Lambda}|$: settling inside the cone, and the unbounded ramp of the integrator outside it.

Sweeping the calibration phase. The algorithm initialization must only satisfy the margin condition (24): for testing, the calibration phase was offset by a controlled $\Delta\varphi$, everything else left unchanged, and the frequency held fixed at 47 Hz so that the phase effect is not confounded with the PLL losing lock once the tone stops being cancelled. Figure 5 and Table 7 collect the result. As predicted from the theoretical analysis, the transition seems to happen at $\pm 90^\circ$; the divergence rate is also influenced by the phase mismatch, with the integrator ramping faster the further outside the cone it is. It should be noted that for a plant phase error close to 90° the convergence is slow, and the residual is worse than the baseline. This may be a hint to some mismatch due to the calibration procedure, yet to be investigated.

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